

N+ - Network security, Maintenance and Troubleshooting procedures

❖ Troubleshooting Tools

[Q-1] Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.

[ANS]

```
C:\Users\HET>ipconfig

Windows IP Configuration


Wireless LAN adapter Local Area Connection* 4:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 5:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter VMware Network Adapter VMnet1:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::54ba:4a02:fe34:ed65%13
    IPv4 Address. . . . . : 192.168.247.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::afab:7e61:830d:9cb9%4
    IPv4 Address. . . . . : 192.168.101.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::387c:745d:bc9c:c1d0%21
    IPv4 Address. . . . . : 192.168.0.101
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.0.1

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
```

[Q-2] Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.

[ANS]

```
C:\Users\HET>ping www.google.com

Pinging www.google.com [142.251.155.119] with 32 bytes of data:
Reply from 142.251.155.119: bytes=32 time=14ms TTL=116
Reply from 142.251.155.119: bytes=32 time=14ms TTL=116
Reply from 142.251.155.119: bytes=32 time=13ms TTL=116
Reply from 142.251.155.119: bytes=32 time=14ms TTL=116

Ping statistics for 142.251.155.119:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 13ms, Maximum = 14ms, Average = 13ms
```

[Q-3] Run the tracert (or traceroute on Linux/Mac) command to www.flipkart.com and list the number of hops it takes to reach the destination.

[ANS]

```
C:\Users\HET>tracert www.flipkart.com

Tracing route to flipkart.com [163.53.76.86]
over a maximum of 30 hops:

  1      1 ms      1 ms      2 ms    192.168.0.1
  2      8 ms      1 ms      1 ms    dsldevice.lan [192.168.1.254]
  3     10 ms      4 ms      2 ms    10.130.192.1
  4      *         *         *       Request timed out.
  5     12 ms     12 ms     12 ms    136.232.112.109
  6      *         *         *       Request timed out.
  7     30 ms     27 ms     28 ms    115.242.187.186
  8      *         *         *       Request timed out.
  9      *         *         *       Request timed out.
 10      *         *         *       Request timed out.
 11      *         *         *       Request timed out.
 12     34 ms     34 ms     33 ms    163.53.76.86

Trace complete.
```

[Q-4] Use the nslookup tool to find the IP address of www.spotify.com and explain in one line what this tool helps you verify.

[ANS]

```
C:\Users\HET>nslookup www.spotify.com
Server: UnKnown
Address: 192.168.0.1

Non-authoritative answer:
Name: atc.spotify.map.fastly.net
Addresses: 2a04:4e42:19::810
           151.101.107.42
Aliases: www.spotify.com
```

➤ One-Line Explanation

- nslookup verifies that a domain name is correctly translated into its IP address by the DNS server.

[Q-5] A friend says they cannot access Zomato's website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check. Hint: Think about checking both network connectivity and DNS resolution.

[ANS]

Step	Command	Purpose
1	ipconfig	Check that the computer has a valid IP address and default gateway.
2	nslookup www.zomato.com	Verify that DNS is resolving the website name to an IP address.
3	ping www.zomato.com	Test whether the website or its server is reachable from the computer.

➤ Explanation

1. **ipconfig** checks whether the network configuration is correct.
2. **nslookup** confirms that DNS is working properly.
3. **ping** tests connectivity and identifies whether the destination server can be reached.